

Foundations of Artificial Intelligence - 2026.1

Sequential Decisions under Risk - Policy Iteration

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Decision Making Situations

Decision Situation Matrix

		Action Effects		
Horizon		Certainty	Risk	Uncertainty
	Atomic	✓	✓	
	Sequential	✓	NOW: 50%	

Class Overview

1. Policy Improvement
2. Policy Iteration with Exact Policy Evaluation
3. Policy Iteration with Iterative Policy Evaluation

State-Action Values

State values assume that one follows the policy π from s on.

It is useful to be more flexible in the *first* move and allow any action $a \in A_s$.

This is expressed in the **state-action value** or **q-value**

$$q^\pi(s, a) := r(s) + \gamma \sum_{s'} \mathbb{P}(s'|s, a) \cdot v^\pi(s')$$

which is the utility of first choosing a in s and thereafter following π .

We can compute q-values from state values (and vice versa since $q^\pi(s, \pi(s)) = v^\pi(s)$).

Policy Iteration

State-Action Values

Once we have state values for policy π , we can compute its q-values.

	1	2	3	4
3	→	→	→	+1
2	↑		↑	-1
1	↑	←	←	←

	1	2	3	4
3	0.812	0.868	0.918	+1
2	0.762		0.660	-1
1	0.705	0.655	0.611	0.388

Determine the four q-values for the initial state (1, 1).

State-Action Values

An alternative way of writing q-values is by the expected value they represent:

$$q^\pi(s, a) = \mathbb{E}^{s, \pi'} [r(S_0) + \gamma v^\pi(S_1)]$$

where π' is any arbitrary policy with $\pi'(s) = a$.

The reason why we can use any arbitrary policy π' is that it is only used for the action in S_0 anyway (after that we use π).

The flexibility to have $\pi'(s) = a$ as the only condition for π' is helpful in some proofs.

Properties of the Policy Iteration Algorithm

Policy Improvement Theorem

For any two policies π and π' with $q^\pi(s, \pi'(s)) \geq v^\pi(s)$ in all states $s \in S$ it holds that

$$v^{\pi'}(s) \geq v^\pi(s)$$

for all states $s \in S$. This is the **policy improvement theorem**.

It means that

- if it is not worse to use π' instead of π once, it is *never* worse in *any* state,
- and therefore we could replace π by π' , and π' is (possibly) an *improvement* over π .

Policy Iteration

The Bellman Equations

Let π' be a policy that is equal to π except for a set $\tilde{S}$ of states where $q^\pi(s, \pi'(s)) > v^\pi(s)$. The policy improvement theorem implies for all $s \in \tilde{S}$ that

$$v^{\pi'}(s) > v^\pi(s)$$

A consequence of this is that no optimal policy π^* can be improved locally, i.e.

$$v^{\pi^*}(s) = \max_a q^{\pi^*}(s, a)$$

for all states $s \in S$.

Policy Iteration

The Bellman Equations

Consequently, for an optimal policy π^* , it necessarily holds in any state s that:

$$v^{\pi^*}(s) = r(s) + \gamma \max_a \sum_{s' \in S} \mathbb{P}(s'|s, a) \cdot v^{\pi^*}(s')$$

This system of $|S|$ equations is often called **Bellman optimality equations**.

Conclusion: π optimal $\Rightarrow \pi$ satisfies the Bellman optimality equation.

Policy Iteration

The Bellman Equations

Interestingly, we also have: π satisfies the Bellman optimality equation $\Rightarrow \pi$ is optimal.

The Bellman optimality equation as an assignment rule constitutes a *contraction*.

This means that there is only one set of state values v that satisfies it.

Since an optimal policy π^* exists and must satisfy it, these state values v must be v^{π^*} .

So if we found some policy π that satisfies the equation, π must be optimal.

Policy Improvement

Equivalent Conditions of Optimality

The following statements are then equivalent:

- π is optimal.
- π satisfies the Bellman optimality equation.
- $v^\pi(s) = \max_a q^\pi(s, a)$ for all s .
- $\pi(s) \in \arg \max_a q^\pi(s, a)$ for all s .

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Policy Iteration

The Advantage Function

Consider a specific policy π and suppose that there is a state s so that

$$v^\pi(s) < \max_a q^\pi(s, a)$$

The policy improvement theorem implies that π is not optimal.

The difference $q^\pi(s, a) - v^\pi(s)$ is called the **advantage** of a in s .

Policy Iteration

Policy Improvement

The **policy improvement** operator exploits this observation and defines:

$$\pi'(s) := \arg \max_a q^\pi(s, a)$$

for all states s for which $v^\pi(s) < \max_a q^\pi(s, a)$.

Since then $q^\pi(s, \pi'(s)) \geq v^\pi(s)$ for all states (and for some even strict), the policy improvement theorem implies that π' is a strict improvement over π .

Policy Iteration

The Policy Iteration Algorithm

The **policy iteration** algorithm simply interleaves policy evaluation and improvement:

1. start with a random policy π
2. compute $v^\pi(s)$ for all s (policy evaluation)
3. compute $\pi'(s)$ based on this (policy improvement)
4. if $\pi = \pi'$, then terminate with π . Otherwise go back to step 2.

Policy Iteration

Termination

How do we know that the algorithm will actually finish and not improve forever?

The Policy Improvement Theorem only leaves room for two cases at the end of each iteration of policy iteration:

1. Either $v^{\pi'}(s) = v^{\pi}(s)$ for all s . But then the q-values do not change, so $\pi' = \pi$.
2. Or $v^{\pi'}(s) > v^{\pi}(s)$ for at least one s .

But in a finite MDP, there exist at most $|A|^{|S|}$ different policies, so this case may happen at most $|A|^{|S|}$ times.

As a result, Policy Iteration is guaranteed to stop in a finite MDP!

Policy Iteration

Early Termination

Importantly, if we stop early, we can still get an interesting optimality bound.

Suppose that the biggest advantage is smaller than ε :

$$\max_a q^\pi(s, a) - v^\pi(s) < \varepsilon$$

Then we are already $\frac{\varepsilon}{1-\gamma}$ -optimal:

$$\|v^* - v^\pi\|_\infty < \frac{\varepsilon}{1-\gamma}$$

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Policy Iteration with Incremental Policy Evaluation

Estimates vs Ground Truth State Values

Incremental Policy Evaluation converges to the true v^π only in the limit.

So we only get *approximations* $\hat{v}^\pi$ and hence also approximations for q-values:

$$\hat{q}^\pi(s, a) = r(s) + \gamma \sum_{s' \in S} \mathbb{P}(s'|s, a) \cdot \hat{v}^\pi(s')$$

Policy improvement will then generally be based on $\hat{q}$, for example,

$$\pi'(s) := \hat{a}_s^* := \arg \max_a \hat{q}^\pi(s, a)$$

Policy Iteration with Incremental Policy Evaluation

If we use greedy policy improvement, we need that

$$q^{\pi}(s, \hat{a}^*) \geq v^{\pi}(s)$$

in order to preserve the requirement of the Policy Improvement Theorem.

The responsibility to guarantee this would be with PE:

Only return estimates $\hat{v}^{\pi}$ if they are exact enough to make sure that the believed best action does not deteriorate current performance.

Policy Iteration with Incremental Policy Evaluation

Even though PE does not know the true v^π and q^π , it can actually guarantee this if:

$$\hat{q}_t^\pi(s, \hat{a}^*) - \hat{v}_t^\pi(s) \geq |\hat{q}_t^\pi(s, \hat{a}^*) - q^\pi(s, \hat{a}^*)| + |\hat{v}_t^\pi(s) - v^\pi(s)|$$

which is the case if for every state s we can verify that

$$\max_a \hat{q}_t^\pi(s, a) - \hat{v}_t^\pi(s) \geq \frac{\gamma(1 + \gamma)}{1 - \gamma} \|\hat{v}_t^\pi - \hat{v}_{t-1}^\pi\|_\infty$$

As long as PE assures this for all states upon termination, PIT usage is safe.

Policy Iteration with Incremental Policy Evaluation

The only remaining undesired case is if PE returns $\hat{v}^\pi$ that suggests that there is *no* advantage in any state even though in reality there would be an improvement possible.

Then the policy would not change and PI would stop pre-maturely with sub-optimal π .

We could require that the inequality would be strict for at least one state. Then we would not stop PE until we have evidence of one strict improvement.

But this yields an infinite loop *if* the current policy is already optimal.

Policy Iteration with Incremental Policy Evaluation

We can resolve both issues by passing a small ε to PE and *add* this stopping criterion:

- $q^\pi(s, a) \geq v^\pi(s) + \varepsilon$ for *some* state s and action a (ensures progress) or
- $q^\pi(s, a) < v^\pi(s) + \varepsilon$ for *all* states s and actions a (ensures termination).

Since provably no action has an advantage of ε or better, we have

$$\|v^* - v^\pi\|_\infty \leq \frac{\varepsilon}{1 - \gamma}$$

if we stop with policy π . In particular, this is the case for s_0 .

Policy Iteration

Policy Iteration Algorithm with Incremental PE

The **policy iteration algorithm with incremental evaluations** interleaves incremental policy evaluation and improvement:

1. start with a random policy π
2. compute $\hat{v}^\pi(s)$ for all s with $\varepsilon > 0$ (incremental ε -certified policy evaluation)
3. compute $\pi'(s)$ based on this (policy improvement)
4. if $\pi = \pi'$, then terminate with π . Otherwise go back to step 2.

The expected utility of the returned policy π deviates by at most $\frac{\varepsilon}{1-\gamma}$ from that of π^* .